RESEARCH PRESENTATION · UK EQUITIES

Is value investing dead—or measured incorrectly?

Reconstructing the evidence for cash-based operating profitability and the new value factor.

Based on Le Quy Duong (2025) · *Journal of Economic Studies*
UK listed equities · July 2000 – June 2024

THE VALUATION QUESTION

CONVENTIONAL VALUE

Book equity / Price

NEW VALUE MEASURE

CBOP / Price

A stronger historical signal is not yet proof of an investable strategy.

The research gap is measurement, not the invention of value investing.

01

BENCHMARK

Fama & French

Book-to-market (B/M) establishes a conventional value signal and HML factor.

02

FOUNDATION

Ball (2016) → Wang (2024)

Accrual-adjusted operating profitability; then a CBOP-to-price strategy tested in US equities.

03

UK EXTENSION

Duong (2025)

Tests portfolio return spreads, NVF factor pricing, and model fit in UK equities.

RQ 1
Has conventional B/M value weakened?

RQ 2
Does CBOP/M better sort UK stock returns?

RQ 3
Does NVF add asset-pricing information?

CBOP changes the numerator—it is not simply cash flow.

ACCOUNTING CONSTRUCT

$$OP = Revenue - COGS - SG\&A$$

$$CBOP = OP + \Delta\ add\text{-}backs - \Delta\ deductions$$

+ Deferred revenue
+ Accrued expenses
+ Accounts payable

− Prepaid expenses
− Inventory
− Receivables

OLD VALUE

$B/M = Book\ equity\ /\ M$

NEW VALUE

$CBOP/M = CBOP\ /\ M$

ILLUSTRATIVE COMPARISON · NOT PAPER DATA

COMPANY A

OP = 20
Δ Receivables = +12
M = 200

4%

CBOP = 8

COMPANY B

OP = 20
Δ Receivables = 0
M = 200

10%

CBOP = 20

Same reported operating profit. Different accrual-adjusted value signals.

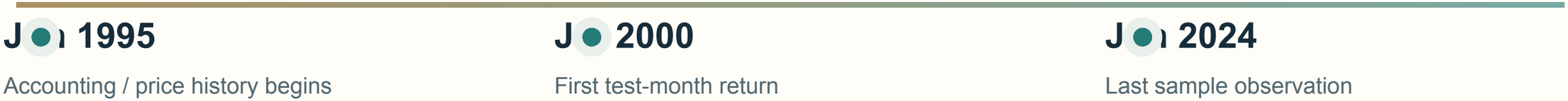
The sample supports a substantial historical test—not yet a real-time replication.

≈1,700
LSE non-financial companies

288
monthly return observations

24 yrs
July 2000 – June 2024

OBSERVATION WINDOW



LSPD · monthly stock returns | **Datastream** · accounting data | Filters: nonfinancial, UK, nonnegative book equity, complete fields.

Publication timestamps unverified

A valuation ratio becomes a strategy through portfolio construction.

INPUT
Prior FY signal
B/M or CBOP/M at the previous fiscal-year end

→

FORM
Each July
Rank eligible stocks into 10 value-weighted deciles

→

HOLD
12 months
Observe monthly returns; next annual rebalance

→

STRATEGY
High − Low
Long top decile, short bottom decile

SEPARATE FACTOR CONSTRUCTION

NVF uses 2 × 3 size / CBOP–M sorts
Not identical to the 10-decile long–short strategy

$$NVF = \tfrac{1}{2}\,(BV + SV - BG - SG)$$

> One accounting signal, two different return constructs. Keep their reported statistics separate.

Four tests answer four distinct questions.

1

Return ordering

Do high-CBOP/M stocks earn higher returns?

DECILE SORTING

2

Risk adjustment

Does alpha persist after controlling for known factors?

TIME-SERIES OLS

3

Incremental pricing

Does NVF add information beyond HML?

SPANNING + FAMA–MACBETH

4

Joint model fit

Do selected models jointly explain test assets?

GRS TEST

RISK-ADJUSTED RETURN REGRESSION

$$R_{H-L,t} = \alpha + \Sigma\, \beta_k\, F_{k,t} + \varepsilon_t$$

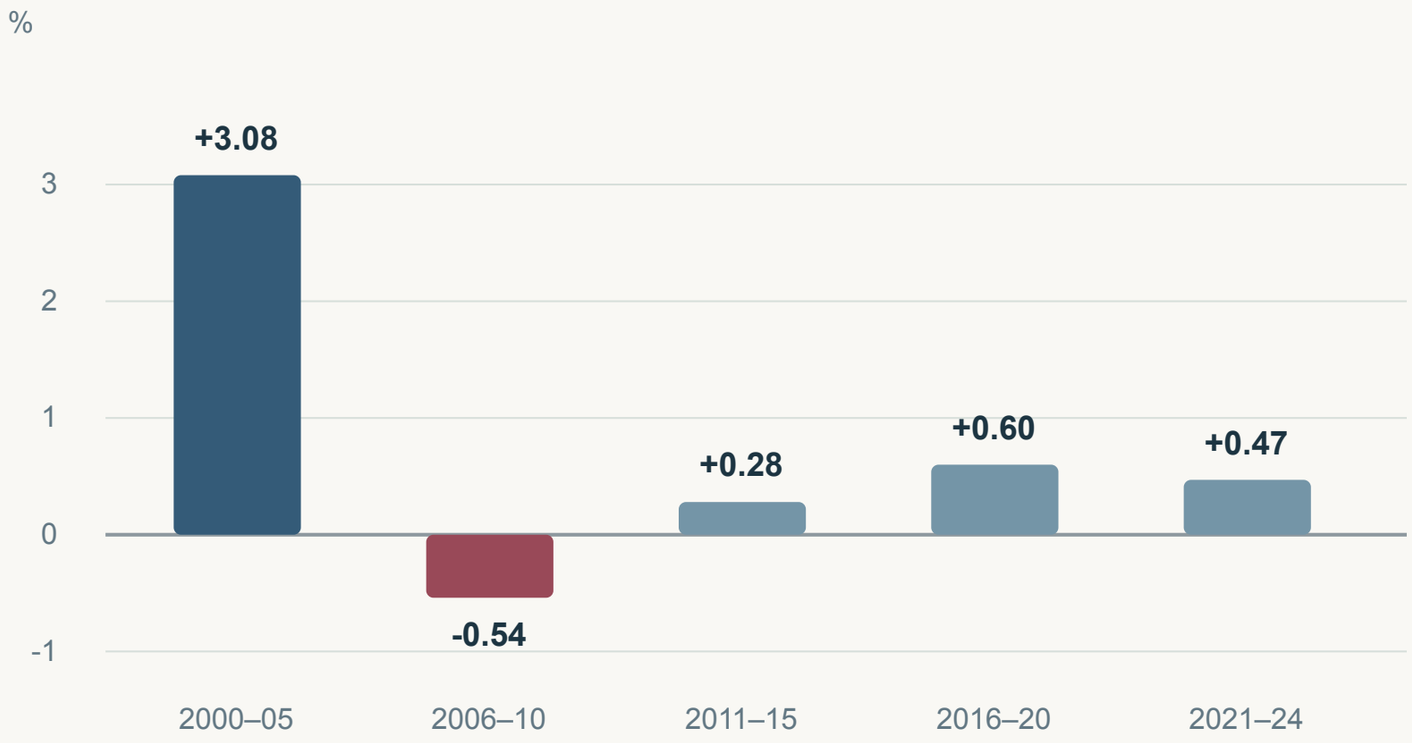
INTERPRETATION

α = return unexplained by *specified* factor controls, not net profit.

Factor controls: CAPM, Fama–French 3/5/6, and Fletcher et al. (2024) UK 5-factor. No IV, GMM, or causal identification claimed.

Conventional B/M value was strongest early—and weak after 2006.

B/M HIGH–LOW RETURN · MONTHLY % · FIVE SUBPERIODS



FULL SAMPLE 2000–2024

+0.790%

B/M gross decile spread per month · **t = 2.312**

RETURN COMPOSITION

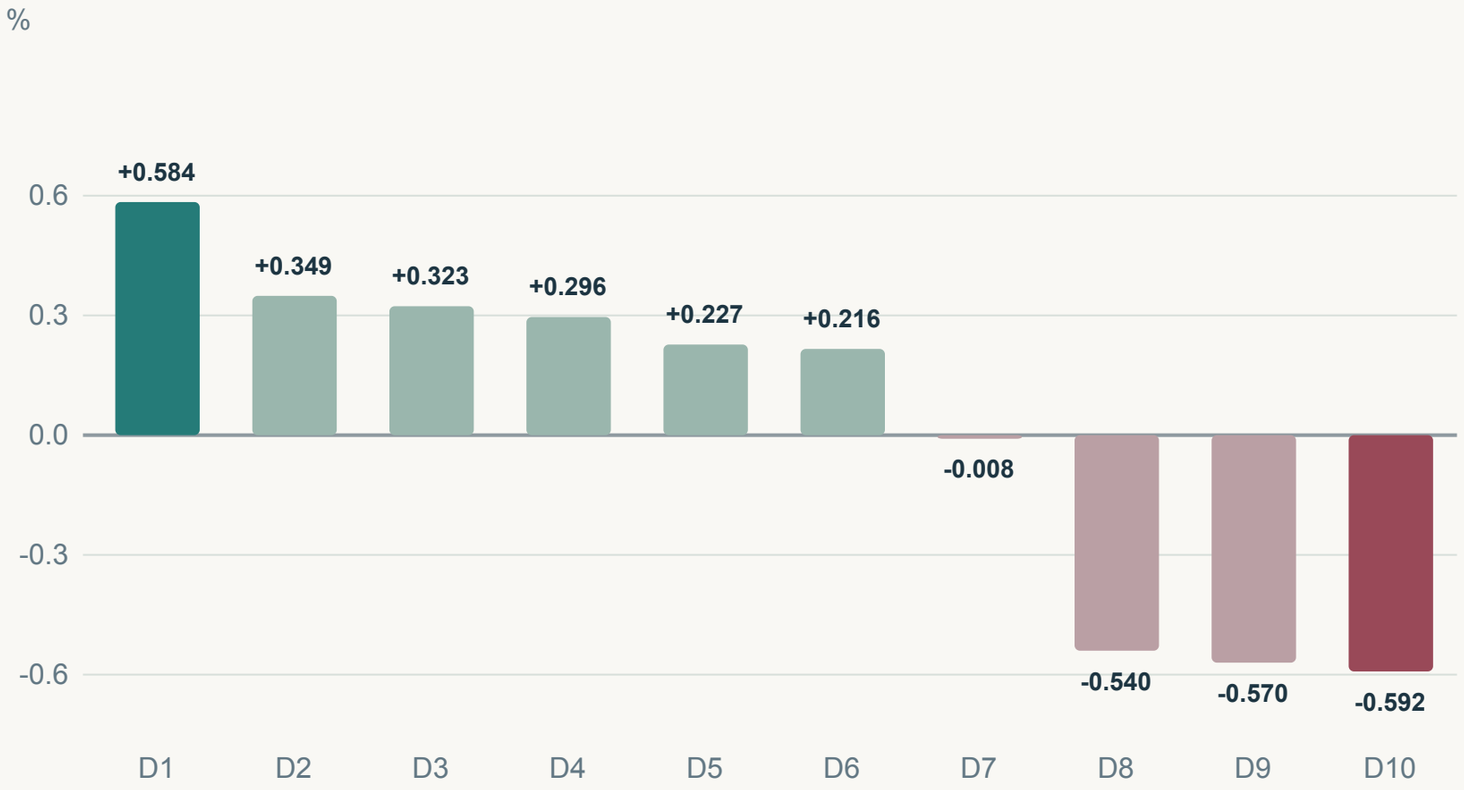
High B/M	**+0.150%**
Low B/M	**−0.640%**

A large share of the spread comes from the short leg.

After 2006: **≈+0.18% per month**. Different time window from full sample.

CBOP/M creates a 1.175% monthly gross spread.

VALUE-WEIGHTED RETURNS · CBOP/M DECILES · % PER MONTH



HIGH–LOW / MONTH

1.175%

t = 3.615

BOTH TAILS CONTRIBUTE

High CBOP/M **+0.584%**

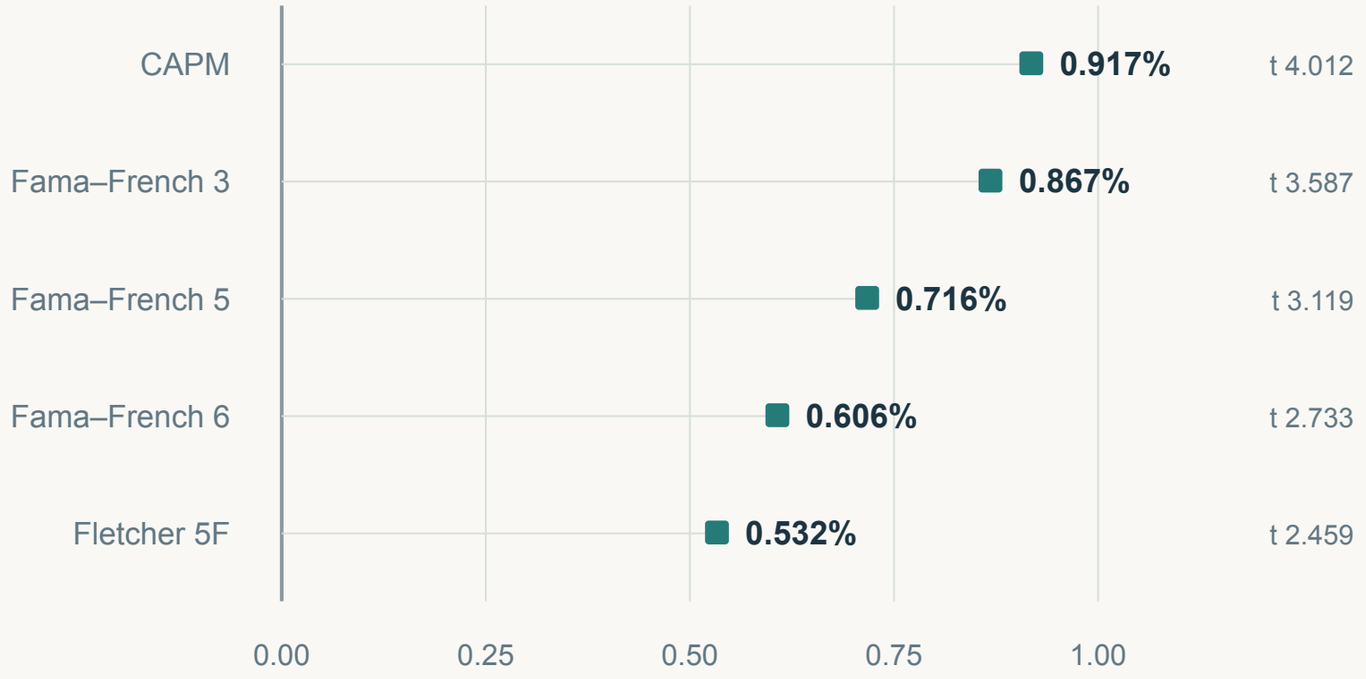
Low CBOP/M **−0.592%**

Matched B/M full-sample spread: **+0.790%**. Difference is descriptive, not separately tested.

Paper Table 6, p. 12 · Gross returns

Positive alpha survives five tested risk models.

HIGH–LOW ALPHA · PERCENTAGE POINTS PER MONTH



MOST DEMANDING BENCHMARK

0.532%

Fletcher 5-factor alpha / month · **t = 2.459**

MODEL EXPLANATORY POWER

R^2 = 0.194

Fletcher factor regression for the strategy (Table 8). Not predictive R^2.

An intercept is not a post-cost trading return or proof of causal mispricing.

Paper Tables 7–8, pp. 13–14

NVF overlaps with HML—yet retains distinct pricing information.

FACTOR OVERLAP

0.461

Correlation of NVF with conventional HML

NVF mean **+0.831%/month**
Different from decile strategy.

SPANNING REGRESSION

+0.358%

NVF alpha / month

t = 2.344

Residual mean return after selected controls.

FAMA–MACBETH · JOINT MODEL

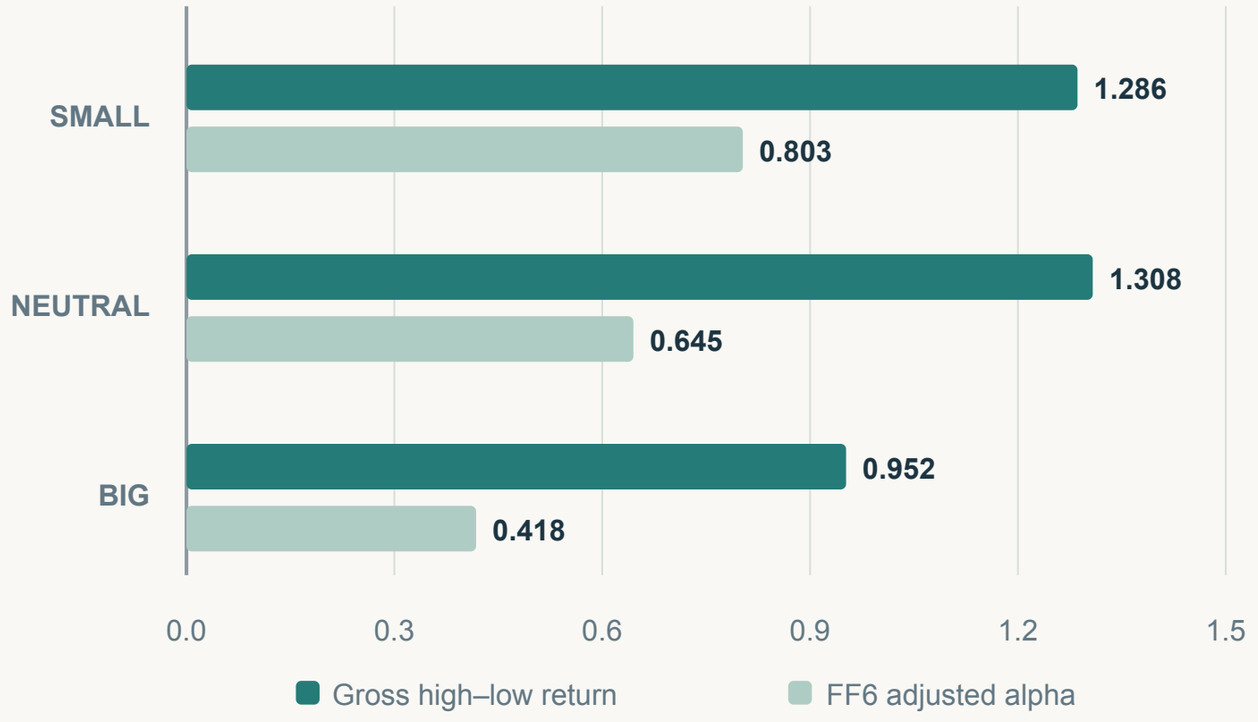
NVF λ	HML λ
0.466%	**0.113%**
t = 4.323	t = 1.576

Prices of factor exposures—not strategy returns.

Different objects: **1.175%** decile strategy · **0.831%** NVF mean · **0.466%** cross-sectional λ NVF.

The CBOP/M spread is not confined to small stocks.

SIZE PARTITIONS · GROSS RETURN VERSUS FF6 ALPHA · % MONTHLY



BIG STOCKS

0.952%

Gross high–low per month · t = 2.953

FF6 ALPHA / BIG

0.418%

Model-adjusted monthly alpha

Market capitalization alone does not establish **liquidity or capacity**.

Replacing HML with NVF improves GRS diagnostics on these test assets.

25 SIZE–B/M TEST PORTFOLIOS · 288 MONTHS

FACTOR MODEL	HML P	NVF P	NVF GRS
FF3	0.006	**0.027**	1.668
FF5	0.017	**0.122**	1.362
FF6	0.092	**0.354**	1.090
Fletcher 5F	0.232	**0.620**	0.923

HML vs NVF versions use corresponding versions of each factor model. Source: Table 12.

NULL HYPOTHESIS

All 25 test-portfolio alphas jointly equal zero.

FLETCHER 5F · P-VALUE

.232 → .620

FF3 with NVF is **still rejected** at 5%: p = 0.027. High p ≠ proof of model truth.

Three evidence boundaries qualify the investment claim.

WHAT THE PAPER SUPPORTS	WHAT REMAINS UNRESOLVED	WHY IT MATTERS
Positive **gross spreads** and factor alphas	Trading, borrowing, and market-impact costs	**Gross signal** ≠ realized net return
Accounts from prior fiscal-year statements	Exact dates each accounting field became public	Publication timing may affect historical ranking
UK sample through June 2024; size and factor tests	Prospective holdout and alternative accounting signals	In-sample fit ≠ universal mechanism

Scientific posture: These boundaries invite stronger experiments; they do not invalidate the historical findings.

Design controlled experiments—before adding new models.

PROPOSED ONLY · NOT PERFORMED

EXPERIMENT 01 · VALIDITY

Point-in-time disclosure

Compare fiscal-year alignment with genuine public release-date information.

Hold constant: CBOP formula, universe, formation rule.
Measure: Spread and alpha.

EXPERIMENT 02 · EXECUTION

Cost-aware investability

Compare gross long–short with net returns under execution, borrowing, and financing costs.

Hold constant: Signals and dates.
Measure: Net return, capacity, drawdown.

EXPERIMENT 03 · MECHANISM

Accounting contribution

Compare B/M, OP/M, and CBOP/M, with industry-neutral robustness.

Hold constant: Eligible stocks and sorts.
Measure: OOS spreads, incremental alpha.

Pre-specify test periods and constraints. Report negative results. No invented backtest curves, risk metrics, or costs.

A better value measure is a research finding. Not yet a trading product.

The next decision is not which machine-learning model to add. It is which assumption to test.

0.790%	1.175%	0.466%
Traditional B/M decile spread · full sample / month	CBOP/M decile spread · full sample / month	Joint-model NVF risk price · Fama–MacBeth / month

RECONSTRUCT → CHALLENGE → TEST

Thank you.

Factor construction and regression notation.

PORTFOLIO FACTORS

$$HML = \tfrac{1}{2}\,(BH + SH - BL - SL)$$

$$NVF = \tfrac{1}{2}\,(BV + SV - BG - SG)$$

$$SMB = \tfrac{1}{3}\,(SMB_{HML} + SMB_{RMW} + SMB_{CMA})$$

B/S = big/small; H/L = high/low; V/G = value/growth in the relevant independent sort.

PRICING AND SPANNING

$$NVF_t = \alpha + \beta' F_t + \varepsilon_t$$

$$R_{i,t} = \lambda_{o,t} + \Sigma\, \lambda_{k,t}\, \hat{\beta}_{i,k} + \eta_{i,t}$$

$$\hat{\lambda}_k = (1/T)\, \Sigma\, \hat{\lambda}_{k,t}$$

β: return exposure; α: conditional intercept; λ: cross-sectional risk price.

> Factor mean, factor alpha, and estimated factor risk price are three different estimands.

A numerical audit must distinguish samples and estimands.

RESULT	REPORTED VALUE	REFERENCE	INTERPRETATION
B/M High−Low, full sample	0.790%; t 2.312	Table 3	Gross decile spread
B/M, 2006–2024	0.18%	Abstract	Different window
CBOP/M High−Low	1.175%; t 3.615	Table 6	Gross decile spread
NVF mean factor return	0.831%	Table 2	2×3 factor return
NVF spanning alpha	0.358%; t 2.344	Table 9	Model-specific alpha
NVF Fama–MacBeth λ	0.466%; t 4.323	Table 10	Cross-sectional price of exposure

Do not subtract a post-2006 B/M return from a full-sample CBOP/M return and report it as an experimental improvement.

Table-level reconciliation questions—not allegations.

QUESTION 1 · SIGN CONVENTION

Printed signs in some t-statistics

Table 7, Fletcher 5F, decile 9: α = −0.186, printed t = +1.539. Table 11 Panel A, Neutral decile 9: return = −0.596, printed t = +1.578.

> Ordinary coefficient / positive standard error should share sign. Verify whether display or transcription errors occurred.

QUESTION 2 · REGRESSION ADDITIVITY

CAPM spread alpha reconciliation

$$0.511 - (-0.494) = 1.005$$

Table 7 reports High−Low α = 0.917, a difference of 0.088 percentage points.

> If identical samples, factors, and OLS definitions apply, intercept additivity is expected. Ask for construction details.

Without security-level data and regression output, these are verification requests—not evidence that the underlying economic finding is invalid.

Defense sheet: five questions worth being ready for.

Why is NVF mean not 1.175%?

2×3 factor versus 10-decile spread. Different portfolio constructions imply different returns.

Does alpha prove profit after costs?

No. Alpha controls selected pricing factors; it does not deduct turnover, borrow, or execution costs.

Is HML now obsolete?

Conditional inferential redundancy in selected UK tests does not imply universal irrelevance.

Does GRS p = .620 validate the model?

It means failing to reject joint zero alphas on the specified 25 test assets—not proof of model truth.

If you could fund one next experiment, which would you choose?

Point-in-time reconstruction followed by cost-aware out-of-sample evaluation: ensure the signal was available when traded and survived realistic implementation assumptions.

GRS is a joint restriction test—not a trading rule.

GIBBONS–ROSS–SHANKEN TEST · SCHEMATIC FORM

$$GRS = \frac{T - N - K}{N} \times \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{F}' \hat{\Omega}^{-1} \bar{F}} \sim F(N, T - N - K)$$

Inputs

T = 288 months; **N = 25** test portfolios; **K** factors; $\hat{\alpha}$ = vector of model alphas; $\hat{\Sigma}$ = residual covariance; $\bar{F}$ = mean factor returns; $\hat{\Omega}$ = factor covariance.

Interpretation

H_0: all test-asset intercepts equal zero. A low p-value is evidence against joint zero alphas, conditional on the model and test assumptions.

Supported: Compared on the same 25 size–B/M portfolios, replacing HML with NVF improves the reported GRS diagnostics.

Not supported: A higher p-value alone proves that a factor model is true or that a trading portfolio earns net alpha.